

How e-mail works

Each day, billions of e-mail messages are sent across the Web through an array of networks and servers. Thanks to the easy-to-use tools that we use to send and receive e-mail, the process from our point of view is pretty transparent. All it takes is an e-mail address and a message. It's so quick and easy that you could probably send over a half dozen mail messages without even realising it. No wonder it is such a popular communication tool.

Ever wonder how that electronic message is sent across the world in just a matter of seconds? How is it that e-mail always lands up in the correct mailbox? The answers to these questions are simpler than you think.

Getting started

Aside from an Internet connection, the only piece of software that you need to install is an e-mail client. If you're already making use of electronic mail, you might have the likes of Outlook Express, Pegasus or similar programs installed on your computer. E-mail messages are similar to letters sent by snail mail and can be broken down into two major parts.

One part of a mail message is known as the header. Similar to an envelope, it contains the name and address of the recipient(s) and the subject of the message. Some clients also show your name, address and date that the message was sent.

The second part of an electronic message is the body. Like a letter which can be stuffed with text and pictures, the body of an e-mail can hold text, images and other various attachments like compressed files.

How it works

There are many computers linked to the Internet which run software applications dedicated to performing a particular task. These servers, as they are called, can either 'serve' Web pages, files or even focus on the task of e-mail processing. Once you've composed a message and clicked the Send button, the e-mail client forwards the message to a server which takes care of all things related to the e-mail's destination.

Both the e-mail client and the mail server communicate using a specialised language designed to handle the mail that you send. This protocol, known as SMTP or Simple Mail Transfer Protocol, is the standard used across the Internet for sending electronic mail. When you send the message, the mail client sends it across to a local SMTP server. Your ISP usually has servers set up for these purposes operating 24 hours a day. For example, VSNL users in Mumbai can use 202.54.1.18 as their SMTP server.



Once your message is sent to the local server, the first step is to look up the recipient's e-mail address stored in the header. Take an e-mail address, somebody@somecompany.com for instance. This address can be

broken down into two parts, the domain, which in this case is somecompany.com and the user name, somebody. If you were to send a mail at this address, your message would be forwarded to your local SMTP server, which in turn would hunt for the somecompany.com domain. Once

found, the message is passed to its destination and stored there until the recipient is ready to download it.

When someone has sent you a message, the SMTP server delivers it to something known as a Post Office Protocol version 3 (POP3) server. Similar to how SMTP is used for sending messages, the POP3 protocol is used to process the act of receiving messages.

Each incoming mail server assigns a certain amount of storage space for every e-mail account on that server. This space, no matter how big or small, is called a mailbox.

When you check your e-mail, the mail software establishes a connection with the mail server that holds your messages. If there are messages waiting, the client will issue a series of commands that begins the download of all correspondence stored on the POP3 server.

As you can see, tracing the path of an e-mail message is much easier than it seems. Anyone in the world with a valid e-mail address can send and receive mail. Almost all ISPs and major online servers offer at least one e-mail address with each account.

Encryption

Unlike letters, electronic mail lacks the element of privacy. Hackers and other would-be intruders can intercept a mail message in transit or directly tap into your mail server and read stored messages from there. Although you use a password to download your messages, the mail server doesn't check to see that the right person typed it in. Naturally, this is a topic of utmost concern.

If you want to keep your confidentiality intact, the way to go about doing it is encryption. This is a process which scrambles ordinary text into coded information which can only be deciphered with the help of keys. With 40-bit encryption, there are billions of possible combinations to decode the information and only one key can work. With 128-bit encryption, there are 300 billion trillion times as many possible keys as 40-bit encryption. This makes it absolutely impossible for an intruder to find the key regardless of how powerful their computers might be.

Security features such as these do not prevent you from browsing through the Web nor does it place any limitations on non-encrypted Web sites. The only thing encryption delivers is security and, of course, peace of mind.

If encrypted mail is an avenue that you wish to pursue, PGP, or Pretty Good Privacy, is the best way to go. PGP is the *de facto* standard for encrypting electronic mail on the Internet simply because it's easy to use, effective, and in most cases, it's free. In order to encrypt a mail using PGP, you need to download the PGP encryption package which is available for download at many sources. Windows and Mac users can download a copy from www.pgp.com.

Keep in mind that if you're sending a PGP encrypted e-mail—the receiving end will need it installed in order to decipher the message. But there shouldn't be any reason to worry since PGP is available across several platforms including OS/2, Amiga and all flavours of Unix.